# Jaesung Lee

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LinkedIn | Google Scholar | Research Page

## Professional Experience

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|---|---|
| ***Assistant Professor***, Industrial and Systems Engineering, Texas A&M University | 2022–present |
| ***Senior Consultant***, Gurobikorea, Seoul, South Korea | 2017 |
| ***Research Engineer/Software Developer***, Diquest Inc., Seoul, South Korea | 2013 – 2016 |

## Education

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|---|---|
| **Ph.D.** Industrial and Systems Engineering, University of Wisconsin–Madison | 2022 |
| **M.S.** Statistics, University of Wisconsin–Madison | 2020 |
| **M.S.** Management Engineering, Korea Advanced Institute of Science and Technology (KAIST) | 2013 |
| **B.S.** Industrial Systems and Information Engineering, Korea University | 2011 |
| **B.E.** Financial Engineering (Interdisciplinary Major), Korea University | 2011 |

## Research Interests

- Domain-informed statistical machine learning and uncertainty quantification
- Bayesian modeling, uncertainty quantification, and data-driven decision-making
- Digital twins and predictive analytics for advanced manufacturing and healthcare

## Publications

### Journal Papers

My advisees are underlined; an asterisk (*) denotes my role as corresponding author.
*Published*

1. S. Aghaee Dabaghan Fard, M. Kim, A. Deep, and **J. Lee**[*]. "Bayesian Joint Model of Multi-Sensor and Failure Event Data for Multi-Mode Failure Prediction," *Technometrics*, pp. 1–15, 2026, https://doi.org/10.1080/00401706.2026.2653564.
    - *Technometrics* is a top journal in science and engineering statistics, with an acceptance rate of approximately 11%.
    - Finalist for the "QCRE Student Paper Competition" at the 2025 IISE Annual Conference & Expo.
    - Invited to present in the *Technometrics* Journal Session at the 2026 Quality and Productivity Research Conference.
2. H. Tian, A. Hanchate, R. Palanna, **J. Lee**, and S. Bukkapatnam. "Explainable Sequential Survival Modeling for Downtime Prognostics in Manufacturing Systems," *Manufacturing Letters*, vol. 49, Supplement, pp. 1286–1297, 2026, https://doi.org/10.1016/j.mfglet.2026.08.035.
3. K. Fu, Z. Li, L. Zhao, A. Deep, **J. Lee**, N. Asadi, and M. Kim. "Examining and mitigating underspecification in neural network-based wafer defect pattern classification," *Journal of Intelligent Manufacturing*, 2026, https://doi.org/10.1007/s10845-026-02890-8.

4. Z. Li, A. Deep, **J. Lee**, and M. Kim. "Manifold-informed Meta-Learning for Out-of-Distribution Detection and Classification in Small-Scale Industrial Datasets," *Journal of Intelligent Manufacturing*, 2026, https://doi.org/10.1007/s10845-026-02873-9.

5. **J. Lee**[*], T. Srikitrungruang, and S. Jahani. "Knowledge-Guided Reinforcement Learning for Preventive Maintenance Planning in Economically Dependent Multi-Component Systems," *IISE Transactions*, vol. 58, no. 11, pp. 1352–1369, 2026, https://doi.org/10.1080/24725854.2025.2610967.

    - Finalist for "2025 QSR Best Refereed Paper Competition" at the 2025 INFORMS Annual Conference.

6. T. Srikitrungruang, S. Aghaee Dabaghan Fard, M. Lemon, **J. Lee**[*], and Y. Zhou. "Robust Physics-Informed Neural Network Approach for Estimating Heterogeneous Elastic Properties from Noisy Displacement Data," *Advanced Science*, vol. 12, no. 48, e08445, 2025, https://doi.org/10.1002/advs.202508445.

    - *Advanced Science* is a premier open access journal with high interdisciplinary impact (IF $> 14$).

7. Z. Li, A. Deep, **J. Lee**, and M. Kim. "A New View of Neural Network-Based Health Index: Connecting Prediction and Operational Decision-Making to Address Underspecification," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 23501–23512, 2025, https://doi.org/10.1109/TASE.2025.3627844.

8. **J. Lee**, S. Zhou, and J. Chen. "Robust Parameter Design on Dual Stochastic Response Models With Constrained Bayesian Optimization," *IEEE Transactions on Automation Science and Engineering*, vol. 21, no. 2, pp. 2039–2050, 2024, https://doi.org/10.1109/TASE.2023.3251973.

9. X. Sui, S. Rangnekar, **J. Lee**, S. Liu, J. Downing, L. Chaney, X. Yan, H. Jang, H. Pu, X. Shi, S. Zhou, M. Hersam, and J. Chen. "Fully Inkjet-Printed, 2D Materials-Based Field-Effect Transistor for Water Sensing," *Advanced Materials Technologies*, vol. 8, no. 22, p. 2301288, 2023 (Back Cover), https://doi.org/10.1002/admt.202301288.

    - Recognized as a top-viewed article of 2023.

10. A. Rahman, T. Rahman, Z. Pei, C. Ufodike, **J. Lee**, and A. Elwany. "Additive Manufacturing Using Agriculturally Derived Biowastes: A Systematic Literature Review," *Bioengineering*, vol. 10, no. 7, p. 845, 2023, https://doi.org/10.3390/bioengineering10070845.

11. C. Huang, **J. Lee**, Y. Zhang, S. Zhou, and J. Tang. "Mixed-Input Bayesian Optimization Method for Structural Damage Diagnosis," *IEEE Transactions on Reliability*, vol. 72, no. 2, pp. 678–691, 2023, https://doi.org/10.1109/TR.2022.3179602.

12. **J. Lee**, C. Wang, X. Sui, S. Zhou, and J. Chen. "Landmark-embedded Gaussian process with applications for functional data modeling," *IISE Transactions*, vol. 54, no. 11, pp. 1033–1046, 2022, https://doi.org/10.1080/24725854.2021.1974129.

    - Featured in the October 2022 issue of ISE Magazine.
    - Awarded as Best Paper in the 2023 IISE Transactions Focus Issue on Design and Manufacturing

13. **J. Lee**, S. Zhou, and J. Chen. "Statistical Modeling and Analysis of k-Layer Coverage of Two-Dimensional Materials in Inkjet Printing Processes," *Technometrics*, vol. 63, no. 3, pp. 410–420, 2021, https://doi.org/10.1080/00401706.2020.1805020.

14. **J. Lee**, J. Son, S. Zhou, and Y. Chen. "Variation source identification in manufacturing processes using Bayesian approach with sparse variance components prior," *IEEE Transactions on Automation Science and Engineering*, vol. 17, no. 3, pp. 1469–1485, 2020, https://doi.org/10.1109/TASE.2019.2959605.

15. Y. Wang, **J. Lee**, and I. Choi. "Indexing by Latent Dirichlet Allocation and an Ensemble Model," *Journal of the Association for Information Science and Technology*, vol. 67, no. 7, pp. 1736–1750, 2016, https://doi.org/10.1002/asi.23444.

*Under Review/Revision*

1. T. Srikitrungruang and **J. Lee***, "Probabilistic Physics-Informed Neural Networks for Estimating Heterogeneous Elastic Properties from Low-Resolution and Noisy Displacement Data," arXiv preprint: https://doi.org/10.48550/arXiv.2607.14563. Major revision at *Technometrics*.
    - Finalist for the 2026 INFORMS QSR Best Student Paper Competition (Ph.D. advisee: Tatthapong Srikitrungruang).
2. S. Aghaee Dabaghan Fard and **J. Lee***, "Scalable Bayesian Deep Kernel Networks via Ensemble-Induced Finite Function Spaces." Under review.
3. A. Salehiyan, **J. Lee**, M. Kim, and A. Deep, "A Scalable POMDP-based Framework for Real-Time Maintenance Planning using Multiple Sensor Signals," TechRxiv preprint: https://doi.org/10.36227/techrxiv.173337778.83986484/v1. Under review.
4. S. Aghaee Dabaghan Fard, M. Kim, A. Deep, and **J. Lee***, "BEIFSurv: Bayesian Ensemble-Induced Finite Function-Space Survival Model for Multi-Sensor Data and Multi-Mode Failure Prediction." Under review.

*Working Papers*

S. Aghaee Dabaghan Fard and **J. Lee***, "Physics-Informed Heterogeneous Bayesian Last Layer Neural Network",

T. Srikitrungruang and **J. Lee***, "Bayesian Physics-Informed Probabilistic Neural Network with Variance-Driven Weighting and Spline Guidance for Inverse Heterogeneous Elasticity Estimation from Low-Resolution and Noisy Data".

### Conference Proceedings

1. **J. Lee**, S. Zhou, and J. Chen. "Sequential Robust Parameter Design With Sample Size Selection". In: *International Manufacturing Science and Engineering Conference*. Vol. 85819. International Manufacturing Science and Engineering Conference. American Society of Mechanical Engineers. 2022, V002T06A024. https://doi.org/10.1115/MSEC2022-85690
2. **J. Lee**, C. Wang, S. Zhou, and J. Chen. "Spatial distribution quantification and control of ink flakes in reduced graphene oxide FET inkjet printing," *Procedia Manufacturing*, vol. 34, pp. 19–25, 2019, https://doi.org/10.1016/j.promfg.2019.06.109.
3. **J. Lee**, D. Kim, K. Kim, and H. Jung, "Efficient compact join algorithm for acyclic conjunctive SPARQL," *ITMSOC Transactions on Information Technology Management*, vol. 1, no. 1, pp. 1–8, 2016.
4. I. Choi and **J. Lee**, "Document indexing by latent Dirichlet allocation," in *proceedings of the 2010 international conference on data mining*, Las Vegas NV, USA, Jul. 12-15, 2010, pp. 409–414, 2010.

## Research Funding

*Awarded funding: $3.08M total ($2.96M federal); $1.94M in federal PI funding.*

- **PI**, National Institutes of Health (NIH)/NIAMS
  **Total:** $1,027,705
  R01, "SCH: Digital Twin Systems for Personalized Prediction of Osteoporotic Spine Fractures Using Physics-Informed Machine Learning," 09/2026–08/2030.
  (Co-Investigators: Jangho Park, Yuxiao Zhou)

- **PI**, National Science Foundation (NSF)
  **Total:** $513,000
  "Trustworthy Digital Twins for Patient-Specific Evaluation of Implant Systems," FDT-BioTech, 06/2027–05/2030.
  (Co-PIs: Yuxiao Zhou, Andrew Read-fuller)
- **PI**, National Institutes of Health (NIH)/NIDCR
  **Total:** $394,886
  R21, "Toward Personalized Dental Surgery Planning: Dental Bone Stiffness Discovery and Strain Prediction Using Physics-Informed Machine Learning," 08/2026–07/2028.
  (M-PI: Yuxiao Zhou; Co-Investigator: Andrew Read-fuller)
- **Senior Personnel**, National Science Foundation (NSF)
  **Total:** $20,000
  "Conference: Collaborative Research at the Intersection of Statistics and Engineering: High-Dimensional Data Analysis," DMS, 01/2026–12/2026.
  (PI: Ardavan Yazdanbakhsh)
- **Co-PI**, Crider Foods
  **Total:** $101,628; **My share:** $36,390
  "Cannery 4.0 Phase 2," 02/2025–11/2025.
  (PI: Satish Bukkapatnam)
  - **My role:** Led the project's failure-prediction analytics, developing machine-learning methods that convert process sensor streams into early warnings for proactive maintenance.
- **Co-PI**, LIFT / U.S. Department of War
  **Total:** $1,000,000; **TAMU portion:** $283,748; **My share:** $93,237
  "VITAL: Virtual Qualification and Certification of an Advanced Structural Material Leveraging Advanced Data-Driven Approaches," AMC 2024, 09/2025–09/2026.
  (PI: Mohsen Taheri Andani; Lead: EOS North America)
  - **My role:** Led the project's data-driven virtual qualification and certification thrust; Mohsen Taheri Andani and industry partners led additive manufacturing and mechanical testing.
  - **Featured article:** LIFT (link)
- **PI**, Texas A&M Institute of Data Science
  **Total:** $20,000
  "Toward Smart Orthopedic Surgery Planning Using Physics-Informed Machine Learning," SPAICD, 01/2025–12/2026.
  (Collaborator: Yuxiao Zhou)
  - **Featured articles:** Dental Tribune (link), Texas A&M Engineering (link)

## Honors

### Awards

- QSR Best Student Paper Competition Finalist (Ph.D. Advisee: Tatthapong Srikitrungruang), INFORMS Annual Meeting, 2026
- 2025–26 Montague–Center for Teaching Excellence (CTE) Scholar, Texas A&M University, 2025
  - University-wide recognition for excellence and innovation in undergraduate teaching (includes a $6,500 teaching grant)
- QSR Best Refereed Paper Competition Finalist, INFORMS Annual Meeting, 2025
- QCRE Student Paper Competition Finalist (Ph.D. Advisee: Sina Aghaee Dabaghan Fard), IISE Annual Conference & Expo, 2025

- IISE Transactions Best Paper Award, IISE, 2024
- Exceptional performance warrant on M.S. exam (statistical consulting exam involving interviewing two clients, writing reports, and orally defending the reports to four faculty committees), Department of Statistics, UW–Madison, 2020
- Best Paper Award for "Efficient Compact Join Algorithm for Acyclic Conjunctive SPARQL," Management and Innovation Technology International Conference, 2015
- Best Master's Thesis Award for "A study on stock price manipulation through spreading false rumor," KAIST, 2013

### Research Recognitions

- Recognized as a top-viewed article of 2023 by Advanced Materials Technologies for "Fully inkjet-printed, 2D materials-based field-effect transistor for water sensing."
- Featured Article in ISE Magazine, IISE, 2022

## Presentations

### Invited Presentations

- "Robust Inference of Unseen Heterogeneous Physical Properties from Imperfect Data: A Unified Probabilistic Physics-Informed Machine Learning Framework," INFORMS QSR Webinar Series, Virtual, October, 2026 (Recording: link)
- "Bayesian Joint Model of Multi-Sensor and Failure Event Data for Multi-Mode Failure Prediction", Quality and Productivity Research Conference (QPRC), Santa Fe, New Mexico, June, 2026
  - (**Invited *Technometrics* journal session**)
- "Domain-Informed Reinforcement Learning for Multi-Component Preventive Maintenance Planning under Economic Dependence", INFORMS Annual Meeting, Atlanta, Georgia, October, 2025
  - (**Invited talk for the QSR Best Refereed Paper Competition**)
- "Toward Smart Reliability Management: Integrating Domain Knowledge with Data-Driven Approaches for Prognostics, Diagnostics, and Decision-Making Using Sensor Data", General Motors, Virtual, July, 2025
- "Domain-informed data-driven methods for prognostics, diagnostics, and decision-making in complex Engineering systems" Virginia Tech, Blacksburg, VA, January, 2025
- "Data Analytics in Manufacturing: Data Collection to Decision Making," Virtual Webinar, The Florida Advanced Technological Education Center (FLATE), Florida, October, 2024
- "Toward Smart Reliability Management: Failure Prediction and Predictive Maintenance Leveraging Condition-Monitoring Signals in Multi-Sensor Systems," Zorich Reliability Workshop, Texas A&M University, September, 2024
- "Data Science Approaches for Modeling and Design with Application in Advanced Manufacturing," Korea University, South Korea, May 2024
- "Landmark-embedded Gaussian process with applications for functional data modeling,", IISE Annual Conference and Expo, Montreal, Quebec, Canada, May 2024
  - (**Invited talk as a recipient of the IISE Transactions Best Paper**)
- "Statistical Modeling of Random Shifting and Shapes in Functional Data and Uncertainty Quantification via Landmark-Embedded Hierarchical Gaussian Processes," Data-Driven Model Reduction, Scientific Frontiers, and Applications Workshop, Texas A&M University, November 2023

- "Statistical Calibration Based on Functional Data with Amplitude and Phase Variations," TAMIDS - Uncertainty Quantification Workshop, Texas A&M University, December 2022
- "Data Science Approaches for Modeling and Design with Application in Advanced Manufacturing," Texas A&M University, March 2022
- "Statistical Modeling and Analysis for Designing and Controlling Advanced Manufacturing," University of Calgary, Virtual, March 2022
- "Data-driven Modeling, Analysis, Design, and Control in Advanced Manufacturing," University of Arkansas, January 2022
- "Data-driven Modeling, Analysis, Design, and Control in Advanced Manufacturing," Korea Advanced Institute of Science and Technology (KAIST), Virtual, October 2021

### Conference Presentations

- "Domain-Informed Reinforcement Learning for Multi-Component Preventive Maintenance Planning under Economic Dependence," INFORMS Annual Meeting, Seattle, WA, USA, October 2024
- "Deep Reinforcement Learning for Maintenance Planning Under Economic Dependency," INFORMS Annual Meeting, Indianapolis, IN, USA, October 2022
- "Sequential Robust Parameter Design With Sample Size Selection", ASME 2022 17th International Manufacturing Science and Engineering Conference, West Lafayette, Indiana, USA, June 2022
- "Constrained Bayesian Optimization on the Robust Parameter Design of Stochastic Functions," INFORMS Annual Meeting, Virtual, October 2021
- "Landmark-embedded Hierarchical Gaussian Process With Applications For Functional Data Calibration,", INFORMS Annual Meeting, Virtual, November 2020
- "Statistical Modeling and Analysis of k-Layer Coverage of Two-Dimensional Materials in Inkjet Printing Processes," INFORMS Annual Meeting, Seattle, WA, USA, October 2019
- "Spatial distribution quantification and control of ink flakes in reduced graphene oxide FET inkjet printing," 47th annual conference on manufacturing research (NAMRC 47), Erie, PA, USA, June 2019
- "Fault Diagnosis in Multistage Manufacturing Processes Based on Bayesian Linear Random Effects Model with Sparse Variance Components Prior," INFORMS Annual Meeting, Phoenix, AZ, USA, November 2018
- "Efficient Compact Join Algorithm for Acyclic Conjunctive SPARQL," Management and Innovation Technology International Conference (MITiCON), Bangkok, Thailand, 2015

## Teaching Experience

**Texas A&M University** College Station, TX

Instructor

- DAEN 210: Uncertainty Modeling (Spring 2024, Spring 2025, Spring 2026)
- ISEN 350: Quality Engineering (Fall 2022, Fall 2023)
- ISEN 614: Advanced Quality Control (Spring 2023, Spring 2024, Spring 2025, Fall 2026)

Course Development

- DAEN 210 Uncertainty Modeling (Summer 2023)

**University of Wisconsin–Madison** Madison, WI

Teaching Assistant

- ISyE 510 Facility Planning (Student Evaluation Score: 4.45/5.0) (Fall 2019)

## Advising

### Ph.D. Advising

- Tatthapong Srikitrungruang, Spring 2023–present (expected Summer 2027)
  - Finalist for the 2026 QSR Best Student Paper Competition at the INFORMS Annual Meeting.
  - Finalist for the 2025 QSR Best Refereed Paper Competition at the INFORMS Annual Meeting.
  - Second place in the ISEN Best Presentation Competition, organized by the INFORMS Student Chapter, 2025.
- Sina Aghaee Dabaghan Fard, Fall 2023–present (expected Summer 2028)
  - Finalist for the QCRE Student Paper Competition at the 2025 IISE Annual Conference & Expo.
- Tanvir Ahmed, Fall 2025–present
  - Runner-up, Student Presentation Competition, INFORMS Student Chapter at Texas A&M University, 2026.

### M.S. Advising

- Noor S. Obeidat, Fall 2024–Fall 2025
  - Outstanding Master of Science Thesis Student Award, 2025.

### Advising for Research Experiences for Undergraduates

- Andres Sanchez (Fall 2026–present; DAEN 491, research credit)
- Gabriel Gonzalez (Spring 2025; admitted to an M.S. program at Harvard University)
- Matthew Lemon (Spring 2025)
- Anna Hartman (Summer 2023)

### Undergraduate Senior Capstone Teams Supervised

- Mallika Parajuli, Sreesh Kanala and Phuong Dao (Company: Lockheed Martin) (Fall 2026)
- Yuval Marom, Graham Miller, Mason Joyner and Nick Mantini (Sponsor: Capco) (Fall 2026)
- Jonathan Harris, Tobin Reynolds and Andreas Sanchez (Sponsor: Navy) (Fall 2026)
- Raymond Gwilliam, Joanna Lin, Gustavo Pinel, Matias Rosenberg Callejas, Vivian Sumbera (Company: Dessert Holdings) (Spring 2026)
- Joseph Zhang, Madeline Claire Fournier, Fahn Saovatharnpong, Skylee McLoud (Company: Merlion) (Spring 2026)
- Jackson T Conant, Callista M D Grass, Ewan Mackie, Michael Tyrrell, Ngan Thanh Pham, Pranav Kulkarni (Company: Applied Materials) (Fall 2024)
- Dave Garrett Mitchell, Margo Angelica, Aidan Steuart, Farrah J Ferguson (Company: Samsung) (Fall 2024)
- Simran Sethi, Julie Victoria Krbashyan, Devon Michael Ramlal, Daniel Levi (Company: Christus Health, St. Patrick's Location) (Spring 2023)
- Aarushi Mudavangatil, Christian Perez, Peter Peng, Isabella Arpero-Bustillos, Saron Bahlbi (Company: Christus Health) (Spring 2023)

- Alexis Zarate, Brayden Mellick, Coleson Mooney, Damaris Cruz Hernandez (Company: Rochester Sensors) (Spring 2023)
- Jennifer Han, Santiago Ruffo, Cole Oliphant, Audrey Jiang (Company: Christus Health) (Spring 2023)
- Jessee Stephens, Aarushi K Agarwal, Brian Nguyen, Brandon Capilla Gaspariano (Company: P&G) (Fall 2022)

## Service

### Professional Service

- Council Member, INFORMS Quality, Statistics, and Reliability (QSR) Section, 2026–2028
- Committee Member, *IISE Transactions* (Design & Manufacturing Focused Issue) 2027 Best Paper Award Committee, 2026
- Organizing Committee Member and Senior Personnel, Workshop on Collaborative Research at the Intersection of Statistics and Engineering (CRSE): High-Dimensional Data Analysis, Wharton School, University of Pennsylvania, Philadelphia, PA, August 11–13, 2026
- Panelist, "Starting Out in Academia" panel in "Doctoral Colloquium", IISE Doctoral Colloquium, IISE Conference, Arlington, TX, 2026
- Associate Editor, Journal of Intelligent Manufacturing, 2024-
- Director of the Korean Research Association for the Business Education, 2023-Present
- Editorial Board Member of the Korean Research Association for the Business Education, 2023-2025

### Institutional Service at Texas A&M University

- Chair, ISEN 350 Course Committee (Fall 2026-), Industrial & Systems Engineering Department, Texas A&M University
- Data Science Focus Area Convener (Spring 2026-), Industrial & Systems Engineering Department, Texas A&M University
- College of Engineering AI/HPC (High-Performance Computing) Working Group (Fall 2025-), Texas A&M University
- ISEN Data Engineering Program Curriculum Committee (Fall 2022-), Texas A&M University
- Member, Faculty Search Committee (Fall 2026), Texas A&M University
- ETAM (Entry to a Major) Review Round 1 (Summer 2026), Industrial & Systems Engineering Department, Texas A&M University
- ETAM (Entry to a Major) Review Round 1 (Summer 2025), Industrial & Systems Engineering Department, Texas A&M University
- Member, ISEN Graduate Affairs Committee (Fall 2023- Spring 2026), Industrial & Systems Engineering Department, Texas A&M University
- Member, ISEN Undergraduate Affairs Committee (Fall 2022- Spring 2023), Industrial & Systems Engineering Department, Texas A&M University
- Judge, ISEN Oral Presentation Competition, Industrial & Systems Engineering Department, Texas A&M University, Oct, 2023

### Proposal Panel and Ad Hoc Review

- Ad hoc Reviewer, Directorate for Technology, Innovation and Partnerships, NSF, Mar, 2025
- Panel, Directorate for Engineering and others, NSF, 2024
- Panel, Directorate for Technology, Innovation and Partnerships, NSF, 2023

### Reviewer/Judge

- Reviewer for the ICQSR 2026 Best Paper Competition, ICQSR, 2026
- Reviewer for the "Full Papers for the IISE QCRE Track", IISE Annual Conference & Expo 2026
- Reviewer for the "IISE-QCRE Best Track Paper Competition", IISE Annual Conference & Expo 2026, May, 2026
- Reviewer for the "QCRE Golomski Award 2025", IISE Quality Control And Reliability Engineering (QCRE) Division, Aug, 2025
- Reviewer for the "INFORMS QSR Best Student Paper Competition", INFORMS Annual Meeting 2025
- Reviewer for the "IISE-QCRE Best Track Paper Competition", IISE Annual Conference & Expo 2025
- Reviewer for the "2024 INFORMS Workshop on Data Mining and Decision Analytics Workshop Best Paper Competition", INFORMS Annual Meeting 2024
- Reviewer for the "2024 INFORMS Data Mining Best Paper Competition", INFORMS Annual Meeting 2024
- Reviewer for the "IISE-QCRE Best Track Paper Competition", IISE Annual Conference & Expo 2024
- Reviewer for the "IISE-DAIS Best Track Paper Competition", IISE Annual Conference & Expo 2024
- Judge for the "QSR Best Student Poster Competition", INFORMS Annual Meeting 2023, Phoenix, AZ, USA, Oct, 2023
- Judge for the "INFORMS Annual Meeting Poster Sessions", INFORMS Annual Meeting 2023, Phoenix, AZ, USA, Oct, 2023
- Reviewer for the "QCRE Golomski Award 2023", IISE Quality Control And Reliability Engineering (QCRE) Division, Aug, 2023
- Reviewer for the "2023 INFORMS Data Mining Best Paper Competition [Student Track]", 2023 Data Mining Best Paper Competition, INFORMS Annual Meeting 2023, Phoenix, AZ, USA, Oct, 2023
- Reviewer for the "IISE-DAIS Best Student Paper Competition", IISE Annual Conference & Expo 2023, New Orleans, LA, USA, May 2023
- Judge of the student competition session "Sustainability in Manufacturing and Cluster Best Student Paper Talks", INFORMS Annual Meeting, Indianapolis, IN, USA, Oct 2022

### Session Chair

- Session Chair, "Domain-Informed Machine Learning in Engineering," INFORMS Annual Meeting, San Francisco, CA, USA, November 2026

- Session Chair of the session "Decision Making on Production System", INFORMS Annual Meeting, Seattle, WA, USA, October 2024
- Session Chair of the session "MSOM/Service Operations Flash Session", INFORMS Annual Meeting, Indianapolis, IN, USA, October 2023
- Session Chair of the session "Advanced Data Analytics for Reliability and Maintenance", INFORMS Annual Meeting, Phoenix, AZ, USA, October 2023
- Session Chair of the session "Data Science & Statistical Learning", 2023 Texas A&M Conference on Energy, September 2023

### Journal and Conference Reviewer

- *IISE Transactions*
- *Technometrics*
- *INFORMS Journal on Computing*
- *INFORMS Journal on Data Science*
- *European Journal of Operational Research*
- *IEEE Transactions on Automation Science and Engineering*
- *Journal of Intelligent Manufacturing*
- *IEEE Transactions on Reliability*
- *Journal of Quality Technology*
- *North American Manufacturing Research Conference (NAMRC)*
- *IEEE Transactions on Semiconductor Manufacturing*
- *Journal of Manufacturing Processes*
- *ASME International Manufacturing Science and Engineering Conference (MSEC)*
- *Reliability Engineering & System Safety*
- *Communications in Statistics: Case Studies, Data Analysis and Applications*

## Patents

### Issued

- **J. Lee**, D. Kim, H. Jang, K. Kim, "Method and apparatus for joining index sequences in SPARQL", Korean Patent KR 101633055B1, granted June 23, 2016, Assignee: Diquest Inc.